

# Visualizing Modeled Social Networks: a Provocation\*

jimi adams, University of South Carolina

2026 September 21

**Abstract:** Network scholarship regularly leverages visualizations. They are thought to enhance readers’ recall, interpretation, and simplification of scholar’s intended takeaways. However, by representing the specifics of single empirical cases, these visualization strategies may not optimize how well they accomplish our scientific purposes. I describe how visualizations often target three non-overlapping sets of aims simultaneously—exploratory description, theoretical conceptualization, *and* empirical interpretation bridging between those. I then highlight how these orientations both necessitate constrain different sets of the varied scientific purposes and principles of visualization. Given these differential demands, this paper provokes the field to consider whether presenting *modeled* network results—in lieu of empirical cases—better satisfies the aims of visualization for particular purposes.

**NOTE:** In preparation for submission to the *Social Networks* theory special issue (which has a submission timeline of Nov 1.)

---

\*This manuscript has benefited greatly from input by seminar participants at the South Carolina Networks Consortium, Duke University’s Social Networks & Health workshop, the Patchwork group at Autonomous University of Barcelona, and Sunbelt 2026, as well as helpful feedback from Jim Moody and David Schaefer. I’m also grateful to Jun Zhao for assisting with materials to include in the illustrative examples. Though they may not all agree with me.

Social network research relies heavily on data visualization (Freeman 2004). These visualizations can serve to characterize, simplify, and focus empirical and/or theoretical claims (Moody et al. 2005; Rawlings et al. 2023). Visualizations are generally more readily digestible than tables of (seemingly endless) coefficients, significance tests, and summary statistics (Healy 2018; Schwabish 2021). Additionally, with the increasing ubiquity of network verbiage in society, they can even convey ideas in ways that require less “translation” work for audiences from outside of academic and research niches (Sayama et al. 2016; Wolff et al. 2026). Quite simply, network visualization is not only common, but can be incredibly useful (Birkett et al. 2021).<sup>1</sup>

Traditionally, this has taken the form of directly visualizing instances(s) of empirically **observed** social networks—whether snapshots of single data waves, or evolving network “movies” for dynamic relational data patterns (Ognyanova 2025).<sup>2</sup> That is, such visualizations are intended to characterize the data *per se*. While this practice predominates, it’s worth asking: (1) what are purposes that motivate why we visualize, and (2) what additional requirements govern appropriate strategies for accomplishing those aims, then (3) whether (there are conditions for which) an alternative strategy serves those *purposes* and *principles* better than the existing approach?

In this paper, my answers to this set of questions suggests many practices of social network visualization are currently misaligned to their purposes—or more precisely, assumes a *single* visualization orientation can successfully address *multiple* sets of aims. As is often the case with split allegiances (Pache and Santos 2010), I contend our current approach could be adjusted to better convey (especially the) *theoretical* takeaways of social networks research. In particular, I contend that an occasional misalignment between these aims, many—but definitely not all—uses of social network visualizations would be better served by visualizing “modeled” networks rather than “observed” network data. I also address how the corollary principles of best practices apply across these differentiated visualization strategies. After making this case following the motivations above, I illustrate this shift in three ways—a trivial example common to how many of us already teach social network modeling, describing several existing strategies that leverage similar properties, and an applied example from a recent research paper.

Before continuing, I should emphasize that when I say “model” I intend this usage in a *theoretical*, not a *statistical* sense. I avoid relegating this point to a footnote to highlight that the strategy I encourage does not apply only to statistically modeled results. Moreover, I should emphasize that this broader applicability in some ways reflects that my proposal in places *imports* practices from non-statistical orientations to network

---

<sup>1</sup>As with any data presentation strategy—visual or otherwise—there are also ample examples of how to do this poorly (e.g., what have come to be known as unintelligible network “hairballs”); I have a few published visuals of my own that I’d *substantially* rework if given the opportunity today.

<sup>2</sup>These typically occur *after* data pre-processing has taken place (e.g., cleaning, recoding, etc.), but no further data reduction steps (i.e., are not the product of data *summaries* or analytic interpretations).

research as much as (or perhaps even more so) than developing a method for statistical applications that can be exported to other use cases. That caveat noted, my illustrations below will primarily draw on statistical applications as a concrete demonstration; but I will include how the approach applies beyond these particularities.<sup>3</sup>

## Why (network) visualization?

So, what are the benefits of network data visualizations? And what are the principles we should follow to best achieve those benefits? Here, I begin from the assumption that SNA visualizations seek to optimize their *scientific utility* (broadly construed), and highlight some shortcomings that have emerged in our common practices, which have weakened our ability to accomplish those aims—particularly by assuming a single visualization strategy can address multiple (sometimes competing) sets of aims.

Many of the benefits for network scholarship mirror those that motivate data visualization in science more broadly, especially regarding communication clarity (Rawlings et al. 2023). A fundamental premise is that data visualization enhances **memory** (Schwabish 2021); this reflects what we know from a number of old adages—including that there’s simply value in beauty (Tufte 2006; Halpern 2015). Moreover, this memory benefit leverages the potential **efficiency** available in data visualization that simply cannot be achieved in other strategies for presenting evidence (Healy 2018).<sup>4</sup> Edward Tufte has prioritized not only using this benefit, but attempting to find ways to optimize it, e.g., via efforts to maximize what he called the “ink-to-information” ratio (Tufte 1983). These memory-oriented enhancements can arise both via improved encoding processes *and* recall—e.g., the simple act of including visualizations leverages the complementarity of multi-modal communication strategies’ differential strengths and weaknesses (e.g., compared to text) (Sweller 1988).

The data we use to represent social processes are almost always inherently high-dimensional, and visualizations provide a means for **simplifying** that complexity in structured, interpretable formats (Wickham et al. 2023).<sup>5</sup> This leads visualizations to not only enhance communicative aims, but also to improve readers’ ability to understand and interpret complex social phenomenon (Schwabish 2021). While statistics can summarize data in principled ways, data visualization rarely simply mimics these other forms and representations of data reduction. Instead, they are also “attention-drawing” efforts that highlight certain features of the

---

<sup>3</sup>Specifically, some of the elements proposed below are directly adopted from other corners of SNA where they are already variably common practices (if labeled differently), and (partially) pave the way for how statistically oriented researchers can implement the approach.

<sup>4</sup>There are many pithy statements about the vast amounts of information that can be conveyed in a single image.

<sup>5</sup>Some contend that this is a *constraint* of data visualization imposed on data presentation strategies (by obfuscating its richness), rather than its benefit (Healy and Moody 2014).

data, while simultaneously deprioritizing others (Leifer 1992).

So, how do network scholars *in particular* prioritize these combating aims? Network scholars especially leverage these dimension reduction approaches by translating summaries of (theoretically) salient *social space* into the approaches for producing visualized representations (Rawlings et al. 2023). For example, one such “attention-drawing” move a research may use would highlight certain relational patterns (e.g., community structure), while deprioritizing others (e.g., homophily).<sup>6</sup>

While such trade-offs may seem generally useful, I argue that how researchers prioritize among these various aims frequently (even if implicitly) differ across different stages of the research process, each of which may leverage visualization as a tool. Given these differentiated aims across research stages, applying a single “typical” approach to network visualization may limit the utility of *visualizations* for the particularized purposes at different stages. That is, my hope is to encourage, more, more-varied, and more tailored uses of visualizations in ways that will afford each instance a greater likelihood of achieving the researchers’ needs. In particular, optimal visualization strategies differ substantially between exploratory (Wickham et al. 2023), conceptual/theoretical (Brady et al. 2020), or explanatory aims (Healy and Moody 2014).

## Multidimensional aims: scientific purposes and principles, across research stages

Yet realizing these common aims in practice requires navigating competing demands arising from multiple dimensions of research. First, I distinguish several distinct scientific **purposes**—the communicative reasons for employing visualization. Second, I identify additional scientific **principles** that govern visualization practices in pursuing those purposes. Third, I differentiate three **stages** of research in which visualizations can prove useful: exploration, conceptualization, and explanation. Together, these dimensions show how particular visualization practices necessarily prioritize different **combinations** of these purposes and principles as their roles shift across the stages of the research process.

### Scientific purposes of network visualization

A first dimension in leveraging the utility of network visualizations concerns the various purposes they may accomplish. Although these purposes principally orient visualization toward distinct goals, they are not fully mutually exclusive (Healy and Moody 2014).

---

<sup>6</sup>For example, to emphasize the changes in political polarization over time in the US Congress, Moody and Mucha (2013) employ increased coarse-graining to highlight *group* size and positioning over time, and juxtapose that against the decreased *individual* positioning and distancing.

*Description* seeks to systematically characterize patterns, such as general tendencies or variation, to capture the state of things as they are as accurately as possible (Schwabish 2021). In networks, this can span multiple levels—including features of nodes, ties, sections within networks, or networks as a whole (Rawlings et al. 2023; Moody et al. 2005).

*Case interpretation* makes sense of the particularities of a case, which can help identify its important features as contrasted against the alternatives that could have been (Ragin and Becker 1992; Collins et al. 2024). Additionally, this provides a lens through which to assess how broadly those observed features should inform our understanding of the wider social setting the case represents (Ayala and Koch 2019).

*Simplification* reduces complexity to focus and attune attention to particular features of a social process, while necessarily de-prioritizing other elements (Healy 2018; Leifer 1992). This process often involves decisions about classificatory boundaries and labeling those distinctions (Wickham et al. 2023; Healy and Moody 2014).

Researchers’ *theoretical interpretations* bring empirical observations and broader theoretical frameworks to bear on one another, establishing “what is this a case of?” (Ragin and Becker 1992). In this use, visualizations help identify what we have learned through the lens of theoretical expectations, including what our data reinforce, fail to support, or suggest requires further refinement (Borgatti and Halgin 2011; Liu et al. 2017).

## Scientific principles for visualization practice

Before turning to how visualizations can optimize these purposes across particular tasks, I describe two additional scientific principles that govern how researchers pursue these ends, which should guide any (network) data visualization.

Network visualizations often present unique *ethical* challenges, particularly concerning potential *confidentiality* and *deductive disclosure* (adams 2019; Ortiz Ruiz et al. 2026). The potential for deductive disclosure is broad in any presentation of multiple variables (Rocher et al. 2019), especially in the “closed population” settings common in network research (but see also Zimmer 2010). Relational data provide additional avenues for serial deductive disclosure (Narayanan and Shmatikov 2009).<sup>7</sup> To this end, the frequent practice of plotting empirical networks from data collection efforts may undermine principles of confidentiality and minimal risk, which should lead us to consider alternative visualization strategies that can leverage visualization’s strengths while minimizing those potential risks (Ortiz Ruiz et al. 2026). These limitations do not preclude

---

<sup>7</sup>I.e., the revealed identity of one node in a graph (e.g., myself) can in turn heighten the capacity to identify others (e.g., my friends).

network researchers from visualizing, and even sharing, data (Pasquale et al. 2025). Rather, they suggest that we need to tailor our approaches in ways that current practices do not necessarily optimize.

While the benefits of open-science principles are not new (Partha and David 1994), researchers have recently (re-)discovered enthusiasm for better implementation in practice (Miguel et al. 2019).<sup>8</sup> *Replicability* therefore raises particularly thorny questions for (network) visualizations.<sup>9</sup> Network visualizations entail a combination of node/tie inclusion and exclusion rules, their respective attribute information, and determinations of the positioning of each.<sup>10</sup> This forces consideration of what scope of information (i.e., which variables for which nodes and which ties) and granularity of presentation (e.g., levels of measurement per observation) we include in visualizations. In particular, visualization’s utility may not require including this information in the full detail available in the data. Additionally, because positions in network visualizations are typically defined relatively (vs. absolutely), algorithmic and procedural details (e.g., random-number-generator seeds) can be important for faithfully recreating visualizations that are substantively or aesthetically comparable (Bender-deMoll and McFarland 2005).

### Network visualization across research stages

While research processes entail many potential stages, three commonly employ visualizations as a primary tool. I therefore next outline a stylized typology of visualization strategies that roughly correspond with different stages of research, and consider how they differentially prioritize the purposes and accommodate the principles described above. To foreshadow, this differentiation leads to a suggested strategy of visualizing *modeled analytic results* as a unique combination that remains under-used in network research—a practice I outline in the following section.

*Observed* network visualizations provide a means for examining the case(s) at hand. These characterizations help researchers become familiar with the nature of their data (Martin 2018), both by describing general node, tie, or network patterns of composition and structure and by identifying potential needs for analysis (Freeman 2000).<sup>11</sup> This stage often prioritizes fidelity to exploring the features in the data, accounting for what they contain (Bender-deMoll and McFarland 2005).

---

<sup>8</sup>In fact, in personal communication, one colleague noted that the “sociogram is . . . one of the purest original forms of open science, [. . . and] replicable data transparency.”

<sup>9</sup>While the existing literature sometimes uses “replicable” and “reproducible” interchangeably, and others distinguishing them, those distinctions have at times been employed in exactly opposite ways (Miguel et al. 2019). For clarity, I use only *replicability* to indicate the ability to recreate a visualization in the same way as the original, whether data-linked or otherwise.

<sup>10</sup>Given the aims of this paper, I focus on the replication of *network visualizations* independently of the numerous additional appropriate considerations for how to implement replicable and reproducible research more broadly (Miguel et al. 2019), including the particular demands of network research (Neal et al. 2024; Pasquale et al. 2025).

<sup>11</sup>For example, this entails the pre-processing steps of variable reclassification or anomalous-case handling, (Wickham et al. 2023).

A primary approach to visualizing network data prioritizes this purpose, shaping researchers’ understanding of patterns in the data and—when included in publications—attempting to do the same for readers. While this is often a “first step” toward examining and presenting one’s case, it does not mean that the visualization directly “reflects” reality. It merely indicates that the data have been obtained, minimally processed, shaped into a coherent presentation strategy, and presented to the reader (Wickham et al. 2023; Healy 2018).

The fundamental steps to generating observed network visualizations are: (1) obtaining the data (through primary methods or secondary sources), (2) pre-processing, cleaning, and organizing the data, (3) reducing to focal elements (relational and nodal attributes), (4) transforming that reduction to visual form, and (5) labeling it. While these necessarily entail some degree of simplification—because visualizations cannot capture the data’s high dimensionality—this stage seeks to minimize simplifications or “rotate” the visualization’s priorities sufficiently to examine data’s complexity by iterating through multiple simplifications, one at a time. If included in published papers, this type of visualization serves a similar purpose to the “descriptive statistics” Table 1 that many papers provide—giving the reader a means to orient to the case at hand with a distilled summary. However, publishing observed network visualizations can lead to visuals that are overly complex in ways that obfuscate what we can learn from them, draw attention to the particularities of the case, rather than the general principles the case is examined to characterize, be difficult to precisely replicate, raising questions about consistency in interpretation, and introduce potential risks that could violate research ethical standards.

Although primarily descriptive, this type of visualization can also bridge to the analytic aims of research, seeding potential analyses for more inductively oriented work or beginning the process of case interpretation in more deductively oriented work. Many of the pre-processing, presentation, and sense-making steps involved at this stage benefit from intimate knowledge of the data (Potterat et al. 2004). Although this knowledge can inform researchers’ work, it may undermine principles of confidentiality and/or deductive disclosure in ways that may not be advisable or permitted in publishable or directly replicable forms (Ortiz Ruiz et al. 2026).

Some fields readily draw on *visualized conceptual models*, to varying degrees of formalization—ranging from those that primarily serve to “organize one’s thoughts” (Brady et al. 2020) to approaches that formally articulate analytic aims and assumptions [e.g., directed acyclic graphs; Elwert (2013)]. Network research has periodically leveraged such visualization aims to specify how *combinations* of features can lead expectations to diverge from various “ideal type” formations of networks. Visualizations can make these configurations concrete in ways that may not be as readily intelligible without them.<sup>12</sup>

---

<sup>12</sup>A very widely used example of this is how “small world” networks, while being very differentiated from random networks,

This form of visualization can explicitly articulate which elements are core to subsequent analyses and which remain unexamined assumptions, thus informing which theorized social processes the analyses evaluate (adams 2020). Conceptual visualizations can formalize the theoretical claims a study evaluates, including the patterns that would support, disconfirm, or suggest alternate interpretations.<sup>13</sup>

In their “pure” form, conceptual visualizations tend to prioritize theoretical interpretation and simplification over all else. Additionally, given the potential for (nearly) “data-free” visualizations of this sort, ethical considerations are likely minimal.

Modeled results<sup>14</sup> visualizations seek to highlight the *theoretical* implications of *interpretation* and/or *explanation* (Rawlings et al. 2023). For example, in social network research with explanatory (e.g., causal) aims, researchers seek to demonstrate how observations in empirical cases (Kadushin 2012) combined with appropriate analytic strategies (Borgatti et al. 2024; McCarty et al. 2019), conform to expectations in ways that support theoretically appropriate interpretations (Borgatti and Lopez-Kidwell 2011). In this case, simplifications tend to highlight theoretical implications over data fidelity (case interpretation). Because an emphasis on theoretical conformity often highlights general patterns or “ideal types,” it need not rely on the specifics of the data—thus allowing researchers to readily maintain confidentiality and facilitate replication (Bender-deMoll and McFarland 2005).

Alternatively, other research may primarily draw on theory and analysis to account for events in particular cases *per se*. In this case simplification and (case) interpretation are still more heavily emphasized than description and full data fidelity. Confidentiality can be maintained through presenting heuristic visualizations—e.g., to elaborate “conceptual generalizability” (Collins et al. 2024)—rather than direct representations of labeled data, also fostering potential replicability.

In sum, different research stages and visualization strategies incorporate different optimizations across the purposes and principles of network visualization. **Table 1** heuristically represents a summary of these potential combinations.

## Implications for visualization practice

As illustrated in the preceding section, visualizations often need to prioritize distinct combinations of the purposes outlined above and may need to adapt in order to adhere to the accompanying visualization

---

require few shortcuts, which does not make them appear particularly statistically distinct from “caveman” graphs, but the difference is readily apparent when visualized (Watts and Strogatz 1998).

<sup>13</sup>Here, visualizations are performing the same clarifying and formalizing task that is accomplished when we employ “writing as thinking” (Zinsser 1993).

<sup>14</sup>As above, “modeled” is not limited to statistical approaches.

**Table 1: Potential Comparative Benefits across Visualization Strategies**

| purpose or principle                        | observed | conceptual | modeled |
|---------------------------------------------|----------|------------|---------|
| describe / explore                          | ✓        | ?          | x       |
| simplification                              | ?        | ✓          | ✓       |
| case interpretation                         | ✓        | x          | ✓       |
| theoretical interpretation                  | x        | ✓          | ✓       |
| confidentiality and/or deductive disclosure | x        | ✓          | ✓       |
| replicability                               | ?        | ✓          | ✓       |

✓ = generally advantageous; ? = context-dependent; x = generally ill-suited.

principles. This potential for non-overlapping combinations can produce sub-optimal visualizations in several distinct ways. Perhaps most obviously, when researchers construct a visualization to serve one of these purposes (e.g., exploratory data analysis), it will not optimally serve other aims (e.g., explanatory) because the combinations necessary to achieve one task well can trade off against the ability to optimize for other purposes.

Perhaps more problematically, when researchers *assume* (or worse, *require*) that visualizations have “universal” benefits, they presuppose that the purposes of data visualization are necessarily shared across this range of uses. One common version of this approach in SNA produces a “descriptive” plot of the observed data and then “overlays” features of the theoretical interpretation of analytic results, in hopes that readers will recognize the “important” signal amid the remaining noise in the data. Unfortunately, this type of compromise strategy can fail to accomplish *any* of our aims well (Pache and Santos 2010). My contention is that network visualizations not infrequently commit one of these two errors.<sup>15</sup> Rather than simply critiquing existing practices, I propose an alternative visualization strategy that optimizes for a combination of purposes not adequately served by our most common approaches, while maintaining the visualization principles detailed above. In particular, I propose that “modeled network visualizations” can more readily combine theoretical interpretation with empirically grounded simplification while preserving confidentiality and facilitating replicability.

## Modeled network visualizations

My fundamental proposition is simple: rather than relying on “observed” networks to present analytic results, we should develop a strategy for presenting “modeled network visualizations.” This is not a groundbreaking suggestion. In fact, many statistical modelers—of social network and other types of data—already

<sup>15</sup>For example, in instances where I have wanted to convey the theoretical takeaways from Adams et al. (2013)—that different types of “risk partnerships” provide different contributions to overall graph connectivity, even while their node-level differences/combinations are minimal—I have found that nearly impossible to readily convey from the (largely exploratory-oriented) visualization we included in the paper—error 1.

enact this practice, but publications do not consistently deploy this approach in ways that fully leverages its potential utility. For example, “eye-balling” consistency of modeled results against observed network visuals—in tandem with formal statistical tests for goodness of fit—is typically taught as a helpful practice in model development and fitting processes (Krivitsky et al. 2023).<sup>16</sup> However, while we use these heuristic network visualizations to improve model fit, authors still often resort to presenting the observed version as their primary network visualization(s) in publications. I have done this multiple times myself.

“Modeled network visualizations” can provide a better alternative when we seek to characterize analytic results in ways that better achieve certain combinations of the aims and principles outlined above.<sup>17</sup> In describing the proposed approach and examples below, I focus on the conditions under which modeled network visualizations offer relative advantages in pursuing this unique combination.

### **Effect plots - a parallel motivation**

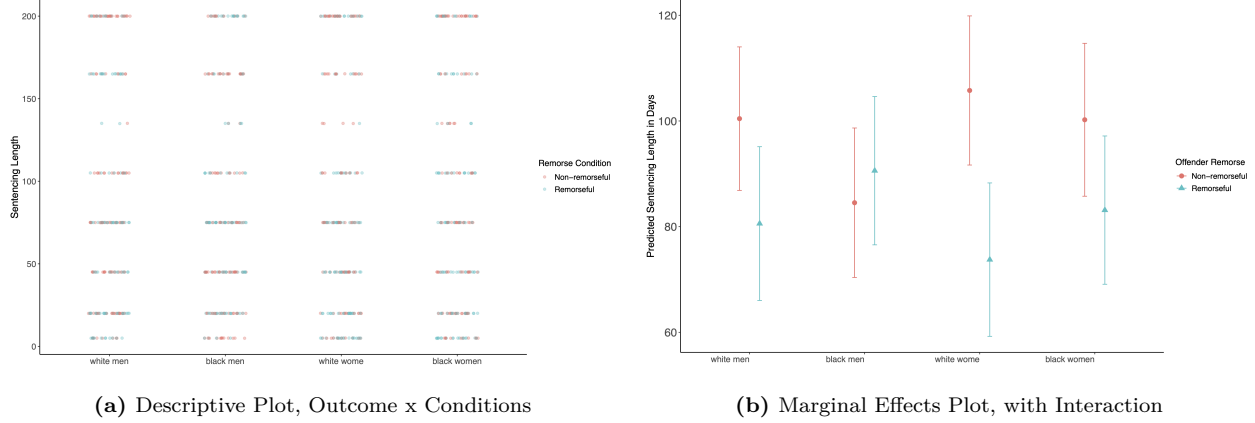
Let me draw a few parallels that illustrate how researchers leverage (parts of) the same logic underpinning this proposed approach when presenting and interpreting modeled results. Logistic-regression coefficients (just like exponential random graph models) are notoriously hard to make substantive sense of on their own (e.g., significance and magnitude of effects are frequently conflated in results interpretations by writers and readers alike). Interaction terms in other forms of regression are similarly difficult to interpret. A common solution has been for researchers to: (1) gather, pre-process, and provide a descriptive summary of their data to give an overview of what the sample being analyzed represents and how key focal features are distributed in the population; (2) conduct statistical analyses to investigate the researchers’ theoretical aims, which are summarized with traditional combinations of betas, standard errors, log-likelihoods, etc. to represent the fit of those claims to empirical estimation; and then (3) transform those modeled results into “predicted probabilities” in ways that help interpret what those modeled results “look like” in their implications (Fox and Andersen 2006).

To “tell the story” that arises from the model results, we do not return to the raw data, provide the individual-level combination of variables for some subset of the population, and rely on those cases’ patterns to convey the implications of our research (in the rare cases that people do, they are often “synthetic” cases, not real ones, e.g., Ayala and Koch 2019). In the case above, the right panel makes apparent that “remorse” benefit in sentencing reductions applies across groups, except for black men (Zhao and Rogalin 2024). Deciphering that result from the left panel would be next to impossible; but we also would rarely attempt to use a

---

<sup>16</sup>They can be especially helpful in identifying theoretical limitations of existing models or enhancing stability of estimates.

<sup>17</sup>I do not also contrast these with “conceptual” visualizations in what follows, because researchers often develop those intentionally in *non-empirical* ways.



**Figure 1:** Predicting Sentencing Length from Expressed Remorse, split by Offender Race & Gender (Zhao & Rogalin 2024).

descriptive plot for those purposes. To do so would not only fail to clearly tell the aggregate story we have learned in the course of theorization and analysis.<sup>18</sup> So why, in network research, do we so often feel uniquely compelled to present the “real” pattern of relationships from our data as a means to interpret and convey our findings in visual form?

### Blockmodels, motifs, & coefficients - partial *network* parallels

It is also important to note that the basis of the general idea—if not the complete practice I am recommending—is already in use in some domains of social network research. Perhaps the most direct corollary is how blockmodeling results tend to be presented as a representations of role profiles (Doreian et al. 2005). The image-matrix and/or reduced-graph presentation distills the overall role structure of an analyzed graph into summary form (Borgatti et al. 2024). A permuted adjacency matrix arranged by density blocks maps that summary onto the full data, but typically does so for the *observed* adjacency matrix rather than for a modeled representation. Other “coarse-graining” visualization strategies for “large” networks commonly take similar parallel approaches (Newman and Girvan 2004).

Another approach to leveraging visualizations to characterize analytic results in theoretically focused ways comes via *network motifs* (Schwarze and Porter 2021). Network motifs identify *repeated* structural patterns across multiple (often, but not exclusively, *personal*) sub-graphs of networks. They present these patterns as abstracted “common forms,” rather than presenting multiple parallel observed networks and then highlighting commonalities across the repeated panels. These aims are similar to those in blockmodels, though network motifs exclusively consider *local* configurations. However, network motifs—and their visual style

<sup>18</sup>It is additionally well known that overly granular descriptive statistics can inadvertently pose deductive disclosure risks, thus leading to “cell size” thresholds for many reporting standards (Rocher et al. 2019).

of presentation—remain more widely adopted in network science than among social network researchers to date (for an exception see Vacca 2020). Other approaches in ego-network research similarly draw on stylized representations to visualize analytic implications, rather than reverting to empirical networks to explain findings [especially when doing so would violate confidentiality; McCarty et al. (2019)].

Such configurational representations have also been employed to assist with interpreting the results of inferential statistical models of social network data by presenting visualizations of micro-patterns rather than coefficient names in results tables [e.g., in stochastic actor-based models; Snijders and Steglich (2026)].

## Implementing modeled network visualizations

Although modeled network visualizations can take many forms,<sup>19</sup> their implementation follows a similar structure across these potential forms. This section elaborates the four common stages of this process. I accompany the description with an illustrative example drawn from a strategy for teaching and model building in exponential random graph models (Krivitsky et al. 2026; Lind 2012). The final section then illustrates the approach through an empirical example.

### Step 1: Characterize the data to be examined

As with the recent resurgence in attention to the importance of *descriptive* statistics before conducting and presenting any multi-variable analyses (Ayala and Koch 2019; Healy and Moody 2014; Wickham et al. 2023), the first step is to provide a preliminary account of the data to be examined. The full details of what this should entail likely depend on the scientific aims and nature of the data to be investigated (e.g., qualitative or quantitative, closed or open population, etc.), but scholars have recently suggested potentially standardizing these to include a variety of features at the node, tie, and network levels (Neal et al. 2024). In some—but, as with my proposal here, *not all*—cases, an observed (network) visualization can enhance this descriptive task. That is, my proposal doesn’t exclude exploratory network visualizations, instead asking us to reconsider if they’re right for our purposes.

The observed network in panel A of **Figure 2** provides a visualization of the extended example I draw on below while developing this strategy—adapted from a widely used example developed by Benjamin Lind (2012). The data the “hookup” network among characters in *Grey’s Anatomy* and will be used in estimating an exponential random graph model (ERGM) and visualize its results.

---

<sup>19</sup>Recall that I use “modeled” in a theoretical and conceptually based sense, not necessarily a statistical one.

## Step 2: Conduct an analysis appropriate to the research question

The second step is to conduct whatever analyses are appropriate to the theoretical aims of the research questions and the nature of the analytic dataset. This step reiterates a fundamental feature of the proposal here: “modeled” results come in many forms. While the illustration here draws from *statistical* models (ERGMs, in particular), the approach could equally be applied to *any* other form of data-analytic results—ranging from similarly statistically oriented models such as SAOMs through to the primary takeaways from grounded-theoretically oriented ethnographic observations (Dominguez and Hollstein 2014).

Building (statistical) models of observed network data can be a complicated process. In developing “well-behaving” models, estimating of goodness of fit can assist with interpreting what the latest model estimates are—and are not—accounting for in the data. Many tutorials for developing such models employ visualizations of a single network instance from intermediate model results, both as a means to intuit how models work when teaching—a necessary step when enhancing model estimation and fit in real analyses (Krivitsky et al. 2026).

Examining the observed network above in light of theoretical goals,<sup>20</sup> suggests a variety of model parameters to investigate. For example, we could fit models that include degree effects; race, gender, or age effects (both as “activity” and “homophily” parameters); or structural features (e.g., local closure rules (adams 2015)).<sup>21</sup>

## Step 3: Compile an appropriate summary of analytic results

Analytic results and interpretations are necessarily reductions of the full details available in our data (adams 2020). Done well, they draw our attention to the scientifically useful features of the data, lead us to appropriate interpretations, and limit the distractions that keep us from achieving those aims—regardless of the particularities of the case at hand (Healy 2017). The third step is therefore to compile an appropriate overview of analytic results in summary form (whether statistical or otherwise).

In Lind’s example, the fitted ERGM supplies this summary, including both model-specification details and corresponding results for the network features it identifies as important.<sup>22</sup> The results elements of the *Grey’s* example include gender *heterophily* and a restriction on isolates.<sup>23</sup>

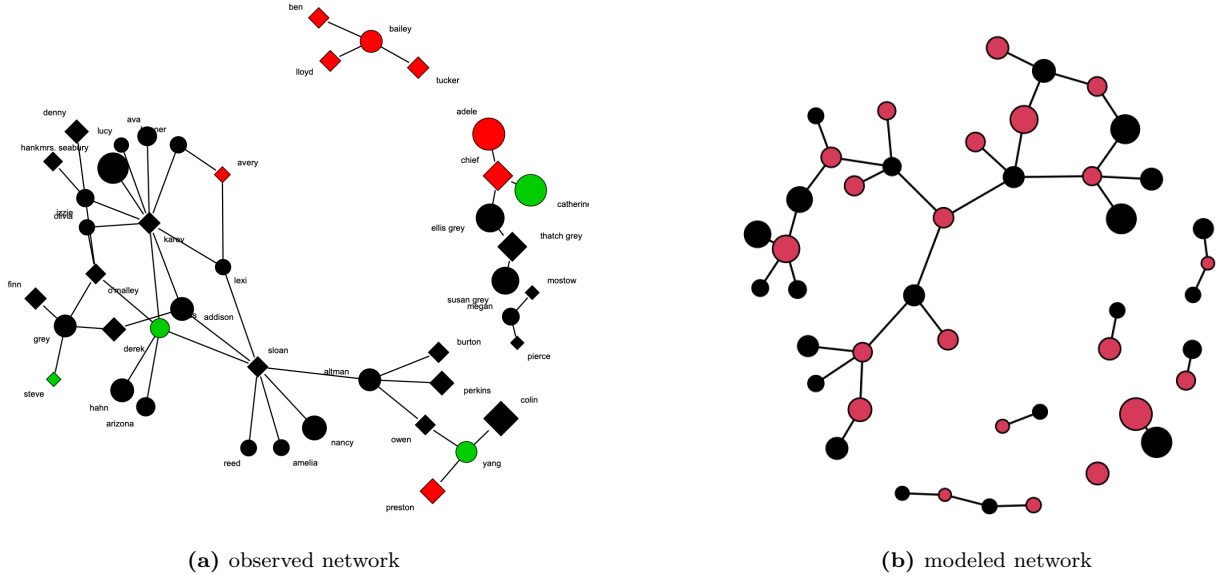
---

<sup>20</sup>In empirical work, this combination likely drives both the data we collect in the first place and the elements we draw from it in exploratory/observed network visualizations.

<sup>21</sup>In **Figure 2** Panel A, “isolates” are not present because they were excluded by definition in the way the data were encoded, so a restriction against isolation should be part of the model.

<sup>22</sup>These should include *all* details necessary to estimate *and* interpret the model. In the case of ERGM, these are returned as part of the model object and include such information as model coefficients and standard errors, goodness-of-fit statistics, and convergence diagnostics (Krivitsky et al. 2023).

<sup>23</sup>This excludes also investigated examples of age or racial homophily. Since formal goodness-of-fit statistics indicated including these produced more poorly performing models, they are excluded from the final results.



**Figure 2: Two visualizations of *Grey's Anatomy* hookup network.** Panel A represents the “observed” data with identities and multiple attributes attached (size=age, color=race, shape=gender). Panel B represents a network generated from the modeled results, including only those exogenous features that are part of the model (node color=gender). This example is adapted from Lind’s (2012) ERGM tutorial.

#### Step 4: Generate a modeled network from that summary

The fourth step is then to convert that summary into a “predicted” network that conforms to the key features of the estimated analytic results. This modeled network visualization will necessarily differ from any observed network visualization by eliminating features that are not “modeled” (Schwabish 2021), reducing the prominence of idiosyncratic variation (i.e., identifying “what is this a case of?” (Ragin and Becker 1992; Collins et al. 2024)), and highlighting only the key elements of the findings that warrant theoretical interpretation (Vertesi 2025). The idea here is to provide an empirically anchored, theoretically prioritized, network visualization that clearly communicates the researchers’ key interpretative takeaways.

In **Figure 2**, panel B visualizes a single draw from the ERGM results for the *Grey's Anatomy* analysis, allowing comparison with the observed network in panel A. The modeled visualization includes only the aspects incorporated in the model, while diminishing the role of idiosyncratic aspects of the data.<sup>24</sup>

Not including names does not diminish the ability to observe important features of the network (e.g., a general, but not absolute, heterosexual partnering pattern). More fundamentally, observed data provide only a single, characterization of reality, invariably including error along with the information they convey.<sup>25</sup>

<sup>24</sup>Here, the visual specifically excludes node labels and nodal attributes of race and age because goodness-of-fit statistics confirmed that these did not contribute to the observed relational pattern. More formal goodness-of-fit tests also confirm the “visual” similarity between the general structures of the two graphs, while other, lesser-performing models showed obvious divergences, visually *and* statistically.

<sup>25</sup>Treating the realized particulars of that instance as central can over-reify what is often only an approximation—though

As such using unlabeled nodes,<sup>26</sup> helps draw our attention to the interpretive implications of our analysis rather than toward the idiosyncrasies of the case at hand (Becker 2017). Furthermore, if these were actual data carrying confidentiality concerns, the modeled visualization—and its rendering of modeled results as unlabeled, “representative” nodes and ties rather than a capture of actual relationships—would also remove risks of sequential deductive disclosure.<sup>27</sup> Moreover, because the visual is based on model results, the model-results object is likely shareable even if the raw data are not,<sup>28</sup> facilitating the replicability of both the results and the visualization.

## An illustrative empirical example

[TO DEVELOP: Introduce a non-confidential empirical case that allows the “revealed” and presumably “un-revealed” versions to appear in tandem, demonstrating why confidentiality concerns can remain consequential even for data that appear non-confidential.]

Here, I will again present a parallel figure that characterizes the “raw” data and then a “modeled result” that conveys the research takeaway.

Work out the aims of visualization more concretely, and elaborate how modeled visualization may do this even better than raw-data visualization can.

[TO DEVELOP: Select an example other than an ERGM application.]

## Quo Vadimus?

As with any visualization strategy, useful examples in the literature are joined by many that are, well, not so much. There are numerous routes to less-informative (network) visualizations that can be avoided by following basic guidelines of visualization—e.g., unintelligible “hairball,” poor layout choices, over-complicate visuals, and other visualization “sins” arise from poor implementation per se (Tufte 1983). However, a similar weakness arise if we assume visualization is a single task with “global optimization” principles. Here, I have sought to make the case that for certain combinations of the purposes and principles outlined above, modeled network visualizations are the better choice.

---

ideally a reasonable one—of the theoretical question being investigated (Becker 2017).

<sup>26</sup>Here we have not merely “removed” labels, but by presenting a modeled network, the nodes are not one-to-one mappings to real data points in any meaningful sense.

<sup>27</sup>For example, given the infrequency of same-sex or inter-racial relationships in the population, even an *unlabeled* version of the graph on the left would allow partners of Derek or “Chief” to likely identify *themselves* in the plot, and thus “discover” their partners having other partners.

<sup>28</sup>It is not uncommon in social network projects to have data-sharing restrictions as part of participation agreements as a means to ensure confidentiality.

I'm fully on board with the fundamental premise that network visualizations one of our field's distinctive calling cards (Freeman 2004), and that this reflects their utility as part of our research toolkit (Freeman 2000). However, this paper also suggests that we've increasingly come to use them as the proverbial hammer, being put to use even when other tools are more suited to the task at hand. As such, I've sought to develop the strategy of model based network visualizations as a tool more attuned to conveying theoretical implications of analytic results, while also potential opportunities for enhanced replicability and confidentiality in their uses.

## References

- adams, jimi. 2015. "Glee's McKinley High: Following Middle America's Sexual Taboos." *Network Science* 3 (02): 293–95. <https://doi.org/doi:10.1017/nws.2015.16>.
- adams, jimi. 2019. *Gathering Social Network Data*. Quantitative Applications in the Social Sciences 180. SAGE.
- adams, jimi. 2020. "What Are COVID-19 Models Modeling?" *The Society Pages*.
- adams, jimi, James Moody, and Martina Morris. 2013. "Sex, Drugs, and Race: How Behaviors Differentially Contribute to the Sexually Transmitted Infection Risk Network Structure." *American Journal of Public Health* 103 (2): 322–29. <https://doi.org/10.2105/ajph.2012.300908>.
- Ayala, Ricardo A., and Tomas F. Koch. 2019. "The Image of Ethnography—Making Sense of the Social Through Images: A Structured Method." *International Journal of Qualitative Methods* 18 (January): 1609406919843014. <https://doi.org/10.1177/1609406919843014>.
- Becker, Howard S. 2017. *Evidence*. University of Chicago Press.
- Bender-deMoll, Skye, and Daniel A McFarland. 2005. "The Art and Science of Dynamic Network Visualization." *Journal of Social Structure* 7: 41.
- Birkett, M., J. Melville, P. Janulis, G. Phillips, N. Contractor, and B. Hogan. 2021. "Network Canvas: Key Decisions in the Design of an Interviewer-Assisted Network Data Collection Software Suite." *Social Networks* 66 (July): 114–24. <https://doi.org/10.1016/j.socnet.2021.02.003>.
- Borgatti, Stephen P, Martin G Everett, Jeffrey C Johnson, and Filip Agneessens. 2024. *Analyzing Social Networks*. SAGE publications Ltd.
- Borgatti, Stephen P., and Daniel S. Halgin. 2011. "On Network Theory." *Organization Science* 22 (5): 1168–81.
- Borgatti, Stephen P., and Virginie Lopez-Kidwell. 2011. "Network Theory." In *Sage Handbook of Social Network Analysis*, edited by John Scott and Peter Carrington. SAGE.
- Brady, Sonya S., Linda Brubaker, Cynthia S. Fok, et al. 2020. "Development of Conceptual Models to Guide Public Health Research, Practice, and Policy: Synthesizing Traditional and Contemporary Paradigms." *Health Promotion Practice* 21 (4): 510–24. <https://doi.org/10.1177/1524839919890869>.

- Collins, Caitlyn, Megan Tobias Neely, and Shamus Khan. 2024. “‘Which Cases Do I Need?’ Constructing Cases and Observations in Qualitative Research.” *Annual Review of Sociology* 50 (1): 21–40. <https://doi.org/10.1146/annurev-soc-031021-035000>.
- Dominguez, Silvia, and Betina Hollstein, eds. 2014. *Mixed Methods Social Networks Research: Design and Applications*. Structural Analysis in the Social Sciences 36. Cambridge University Press.
- Doreian, P., V. Batagelj, and A. Ferligoj. 2005. *Generalized Blockmodeling*. Structural Analysis in the Social Sciences. Cambridge University Press.
- Elwert, Felix. 2013. “Graphical Causal Models.” In *Handbook of Causal Analysis for Social Research*. Springer.
- Fox, John, and Robert Andersen. 2006. “Effect Displays for Multinomial and Proportional-Odds Logit Models.” *Sociological Methodology* 36 (1): 225–55. <https://doi.org/10.1111/j.1467-9531.2006.00180.x>.
- Freeman, Linton C. 2000. “Visualizing Social Networks.” *Journal of Social Structure* 1 (1).
- Freeman, Linton C. 2004. *The Development of Social Network Analysis: A Study in the Sociology of Science*. Empirical Press.
- Halpern, Orit. 2015. *Beautiful Data: A History of Vision and Reason Since 1945*. Duke University Press.
- Healy, Kieran. 2017. “Fuck Nuance.” *Sociological Theory* 35 (2): 118–27. <https://doi.org/10.1177/0735275117709046>.
- Healy, Kieran. 2018. *Data Visualization: A Practical Introduction*. Princeton University Press.
- Healy, Kieran, and James Moody. 2014. “Data Visualization in Sociology.” *Annual Review of Sociology* 40 (1): 105–28. <https://doi.org/10.1146/annurev-soc-071312-145551>.
- Kadushin, Charles. 2012. *Understanding Social Network Analysis: Theories, Concepts & Findings*. Oxford University Press.
- Krivitsky, Pavel N., David R. Hunter, Martina Morris, and Chad Klumb. 2023. “Ergm 4: New Features for Analyzing Exponential-Family Random Graph Models.” *Journal of Statistical Software* 105 (6). <https://doi.org/10.18637/jss.v105.i06>.
- Krivitsky, Pavel N., Martina Morris, Mark S. Handcock, et al. 2026. *Exponential Random Graph Models (ERGMs) Using Statnet*. [https://statnet.org/workshop-ergm/ergm\\_tutorial.html](https://statnet.org/workshop-ergm/ergm_tutorial.html).
- Leifer, Eric M. 1992. “Denying the Data: Learning from the Accomplished Sciences.” *Sociological Forum* 7 (2): 283–99. <https://doi.org/10.1007/BF01125044>.
- Lind, Benjamin. 2012. “Lessons on Exponential Random Graph Modeling from Grey’s Anatomy Hook-Ups.” In *Bad Hessian*. <https://badhessian.org/2012/09/lessons-on-exponential-random-graph-modeling-from-greys-anatomy-hook-ups/>.
- Liu, Wenlin, Anupreet Sidhu, Amanda M. Beacom, and Thomas W. Valente. 2017. “Social Network Theory.” In *The International Encyclopedia of Media Effects*, 1st ed., edited by Patrick Rössler, Cynthia A. Hoffner, and Liesbet Zoonen. Wiley. <https://doi.org/10.1002/9781118783764.wbieme0092>.
- Martin, John Levi. 2018. *Thinking Through Statistics*. University of Chicago Press.

- McCarty, Christopher, Miranda J. Lubbers, Raffaele Vacca, and José Luis Molina. 2019. *Conducting Personal Network Research: A Practical Guide*. Methodology in the Social Sciences Series. Guilford Publications.
- Miguel, Edward, Garret Christensen, and Jeremy Freese. 2019. *Transparent and Reproducible Social Science Research: How to Do Open Science*. University of California Press.
- Moody, James, Daniel A. McFarland, and Skye Bender-DeMoll. 2005. “Dynamic Network Visualization.” *American Journal of Sociology* 110 (4): 1206–41.
- Moody, James, and Peter J. Mucha. 2013. “Portrait of Political Party Polarization.” *Network Science* 1 (1): 119–21. <https://doi.org/10.1017/nws.2012.3>.
- Narayanan, Arvind, and Vitaly Shmatikov. 2009. “De-Anonymizing Social Networks.” *2009 30th IEEE Symposium on Security and Privacy*, May, 173–87. <https://doi.org/10.1109/SP.2009.22>.
- Neal, Zachary P., Zack W. Almquist, James Bagrow, et al. 2024. “Recommendations for Sharing Network Data and Materials.” *Network Science* 12 (4): 404–17. <https://doi.org/10.1017/nws.2024.16>.
- Newman, Mark E., and M. Girvan. 2004. “Finding and Evaluating Community Structure in Networks.” *Physical Review E*. 69: no.026113.
- Ognyanova, Katherine. 2025. *Network Visualization with R*. [www.kateto.net/network-visualization](http://www.kateto.net/network-visualization).
- Ortiz Ruiz, Francisca, Paola Tubaro, José-Luis Molina, et al. 2026. “Recommendations for Conducting Ethical Network Research.” *Network Science* 14 (e17): 1–11.
- Pache, Anne-Claire, and Filipe Santos. 2010. “When Worlds Collide: The Internal Dynamics of Organizational Responses to Conflicting Institutional Demands.” *Academy of Management Review* 35 (3): 455–76.
- Partha, Dasgupta, and Paul A David. 1994. “Toward a New Economics of Science.” *Research Policy* 23 (5): 487–521.
- Pasquale, Dana K., Tom Wolff, Gabriel Varela, et al. 2025. “Considerations for Social Networks and Health Data Sharing: An Overview.” *Annals of Epidemiology* 102 (February): 28–35. <https://doi.org/10.1016/j.annepidem.2024.12.014>.
- Potterat, J. J., D. E. Woodhouse, S. Q. Muth, et al. 2004. “Network Dynamism: History and Lessons of the Colorado Springs Study.” In *Network Epidemiology: A Handbook for Survey Design and Data Collection*, edited by Martina Morris. Oxford University Press.
- Ragin, Charles C., and Howard S. Becker. 1992. *What Is a Case?: Exploring the Foundations of Social Inquiry*. Cambridge University Press.
- Rawlings, Craig M., Jeffrey A. Smith, James Moody, and Daniel A. McFarland, eds. 2023. “How Are Social Network Data Visualized?” In *Network Analysis: Integrating Social Network Theory, Method, and Application with R*. Structural Analysis in the Social Sciences. Cambridge University Press. <https://doi.org/10.1017/9781139794985.006>.
- Rocher, Luc, Julien M. Hendrickx, and Yves-Alexandre de Montjoye. 2019. “Estimating the Success of Re-Identifications in Incomplete Datasets Using Generative Models.” *Nature Communications* 10 (1): 3069. <https://doi.org/10.1038/s41467-019-10933-3>.

- Sayama, Hiroki, Catherine Cramer, Mason A. Porter, Lori Sheetz, and Stephen Uzzo. 2016. “What Are Essential Concepts about Networks?” *Journal of Complex Networks* 4 (3): 457–74. <https://doi.org/10.1093/comnet/cnv028>.
- Schwabish, Jonathan. 2021. *Better Data Visualizations: A Guide for Scholars, Researchers, and Wonks*. Columbia University Press.
- Schwarze, Alice C., and Mason A. Porter. 2021. “Motifs for Processes on Networks.” *SIAM Journal on Applied Dynamical Systems* 20 (4): 2516–57. <https://doi.org/10.1137/20M1361602>.
- Snijders, Tom A. B., and Christian Steglich. 2026. *Stochastic Actor-Oriented Models for Longitudinal Networks*. Structural Analysis in the Social Sciences. Cambridge University Press.
- Sweller, John. 1988. “Cognitive Load During Problem Solving: Effects on Learning.” *Cognitive Science* 12 (2): 257–85. [https://doi.org/10.1016/0364-0213\(88\)90023-7](https://doi.org/10.1016/0364-0213(88)90023-7).
- Tufte, Edward R. 2006. *Beautiful Evidence*. Vol. 1. Graphics Press Cheshire, CT.
- Tufte, Edward R. 1983. *The Visual Display of Quantitative Information*. Graphics Press.
- Vacca, Raffaele. 2020. “Structure in Personal Networks: Constructing and Comparing Typologies.” *Network Science* 8 (2): 142–67. <https://doi.org/10.1017/nws.2019.29>.
- Vertesi, Janet. 2025. “Theory-Laden Data Visualization, Drawing-As and Seeing-As in Sociology and in Data Science.” *The American Sociologist* 56 (June): 440–54. <https://doi.org/10.1007/s12108-025-09658-2>.
- Watts, Duncan J, and Steven H Strogatz. 1998. “Collective Dynamics of ‘Small-World’ Networks.” *Nature* 393: 440–42.
- Wickham, Hadley, Mine Cetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd ed. O’Reilly.
- Wolff, Tom, Gabriel Varela, and Jonathan Eastwood. 2026. “Building Computational Literacy in Undergraduate Sociology Using Software: An Example in Teaching Social Network Analysis.” *Teaching Sociology*, ahead of print. <https://doi.org/10.1177/0092055X261422479>.
- Zhao, Jun, and Christabel L. Rogalin. 2024. “Discrimination in Sentencing: Showing Remorse and the Intersection of Race and Gender.” *Social Psychology Quarterly* 87 (4): 309–32. <https://doi.org/10.1177/01902725231219690>.
- Zimmer, Michael. 2010. “‘But the Data Is Already Public’: On the Ethics of Research in Facebook.” *Ethics and Information Technology* 12 (4): 313–25. <https://doi.org/10.1007/s10676-010-9227-5>.
- Zinsser, William. 1993. *Writing to Learn*. Harper Collins.